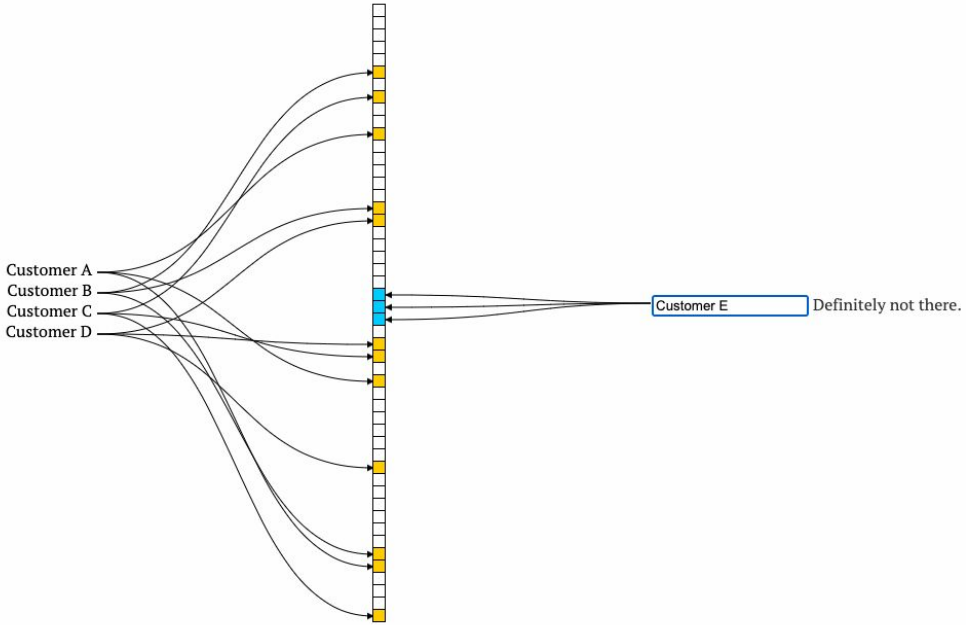
Bloom Filters

Burton Howard Bloom



Visualiser:
https://www.jasondavies.com/bloomfilter/

1. Bloom, Burton H. (1970-07-01). "Space/time trade-offs in hash coding with allowable errors". *Communications of the ACM*. **13** (7): 422–426. doi:10.1145/362686.362692. ISSN 0001-0782.
2. Bloom, Burton H. (1969-08-26). "Some techniques and trade-offs affecting large data base retrieval times". ACM: 83–95. doi:10.1145/800195.805920.

Why bloom filters are useful

- Can tell if an element is **definitely not** in a Bloom.
- Can tell if an element is in a Bloom with some level of probability.
- Deterministic
- Can not enumerate elements
- Can not remove a single element

*Time Complexity : Add, Query: **O(1)***
*Space Complexity: **O(n)** space.*

Maths

These equations provide the optimal answers for the questions:

- How big should the filter be to hold `n` elements with false probability rate `p`?
- How many hashes `k` should we use to achieve that?

$m = -\frac{n \ln p}{\ln^2 2}$	m – filter size (bits) p – error probability (false positive) n – maximum cardinality (capacity)
$k = \frac{m}{n} \ln 2$	k – number of hash functions m – filter size (bits) n – maximum cardinality (capacity)